

Air University



Network Security (Lab)

Year 2026

Mr. Haseeb Ahmed

Academic Year: 2024 – 2028

Department of Cyber Security

USAMA ARSHAD | 247141

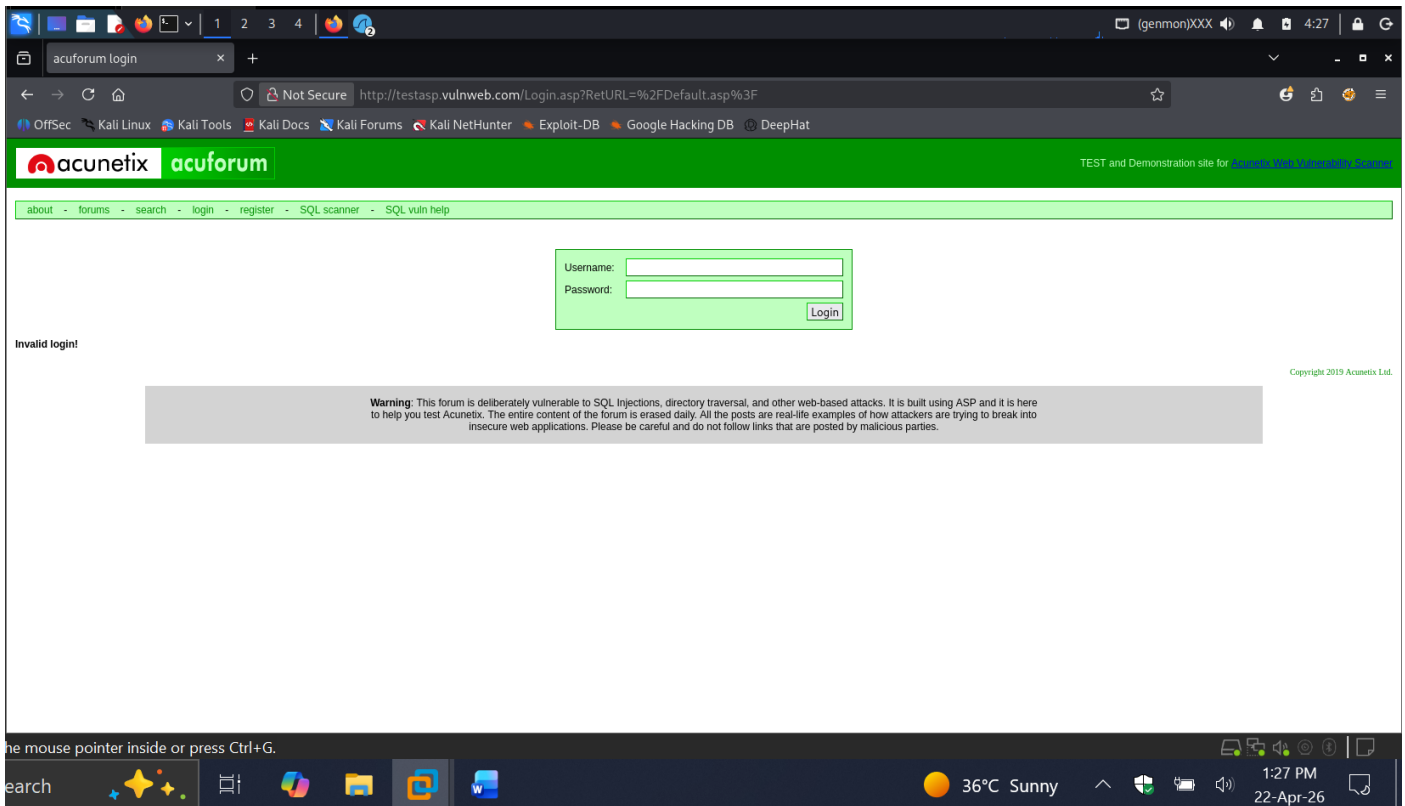
BS – Cyber Security – Fall – 24 (A)

Date of Submission: April 22, 2026

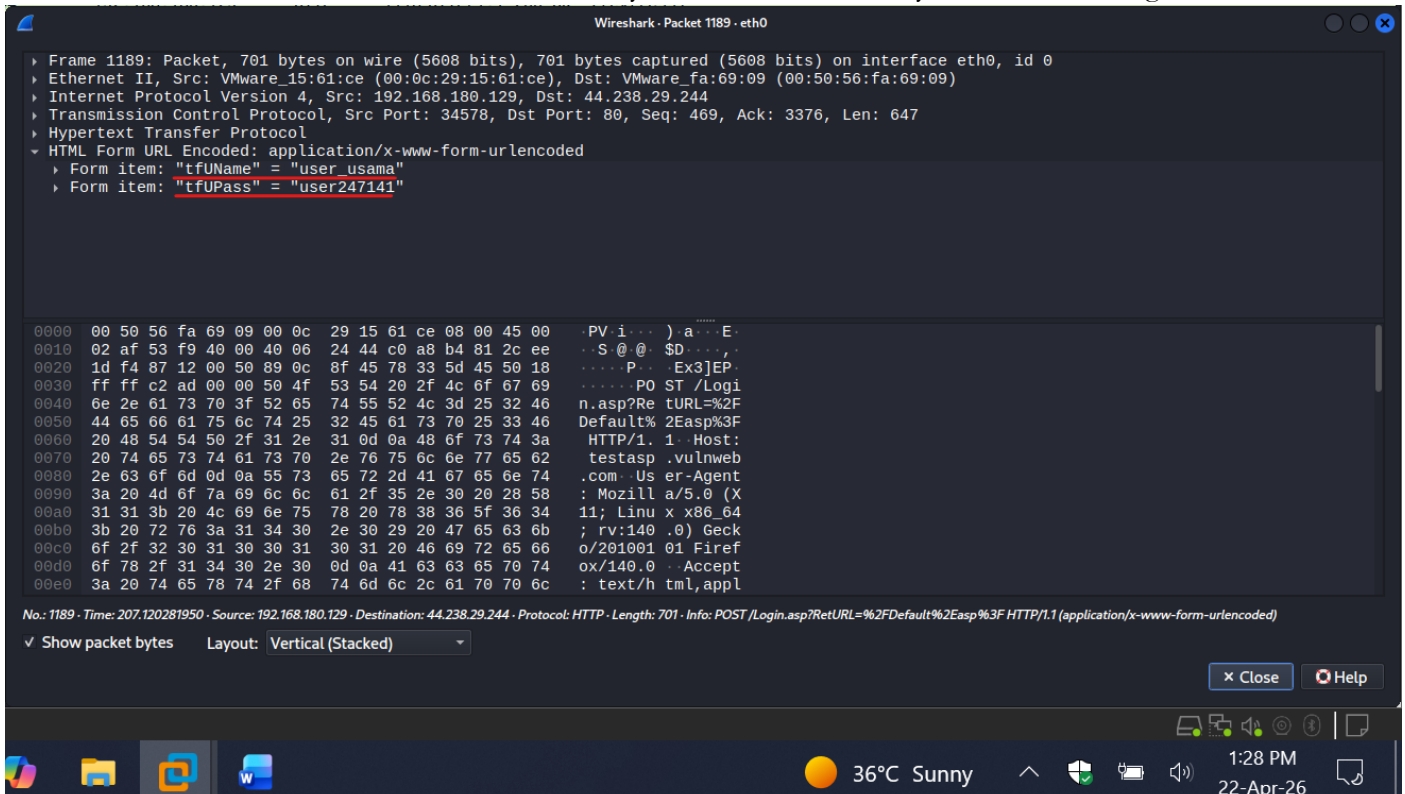
Lab # 12

Wireshark

Website: attempting to log in to an insecure website using the unencrypted HTTP protocol.

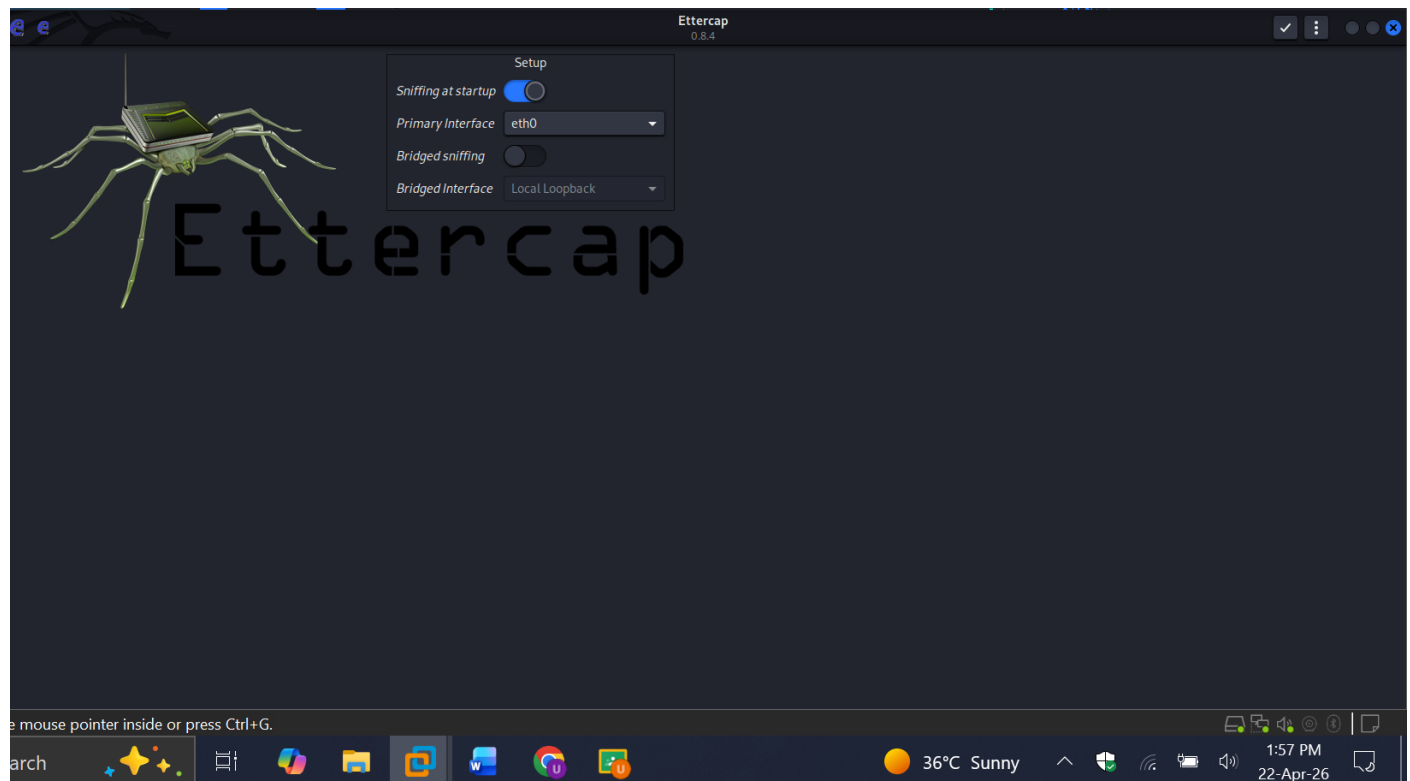


Wireshark: Use Wireshark to reveal that the login credentials, including the username and password, were transmitted in cleartext and are fully visible to anyone monitoring the network.

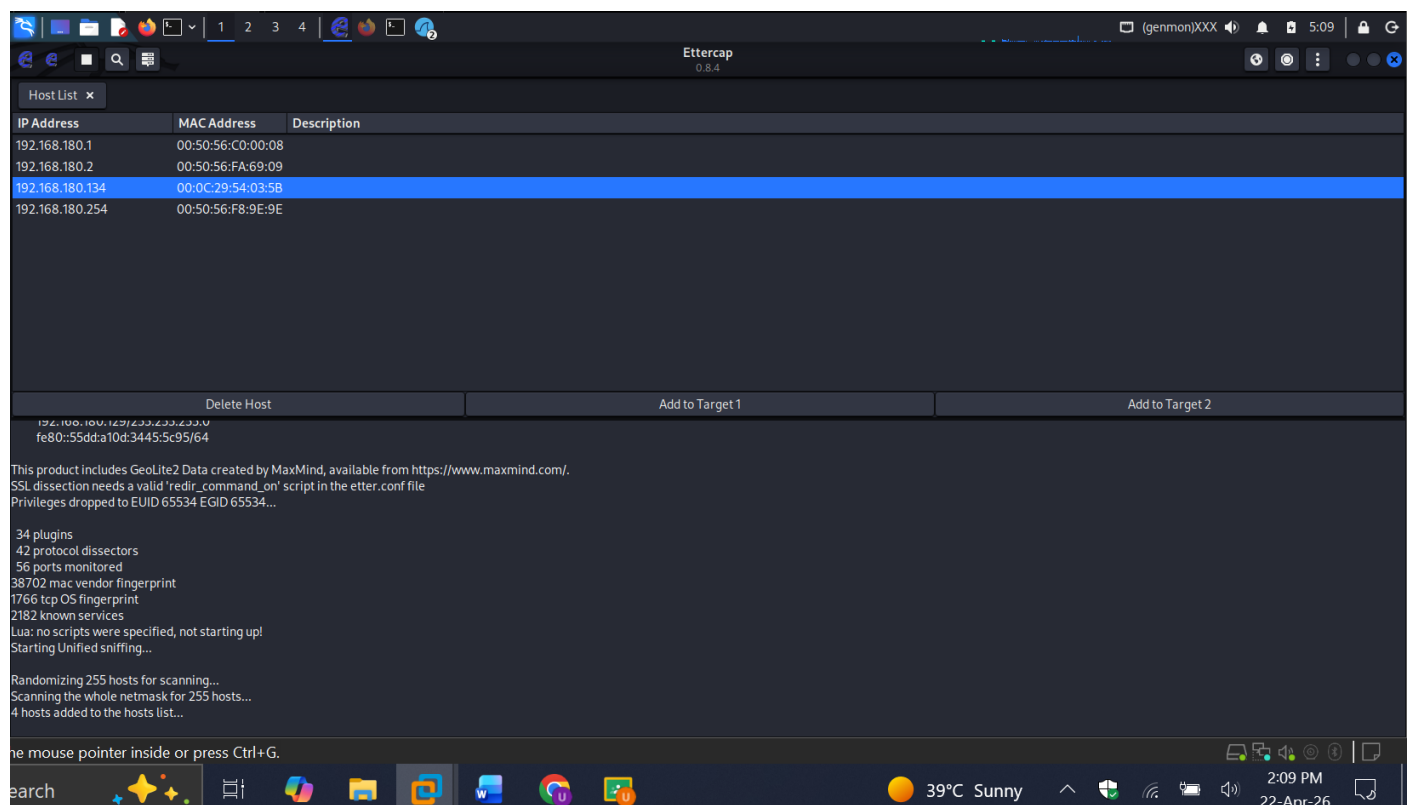


Ettercap

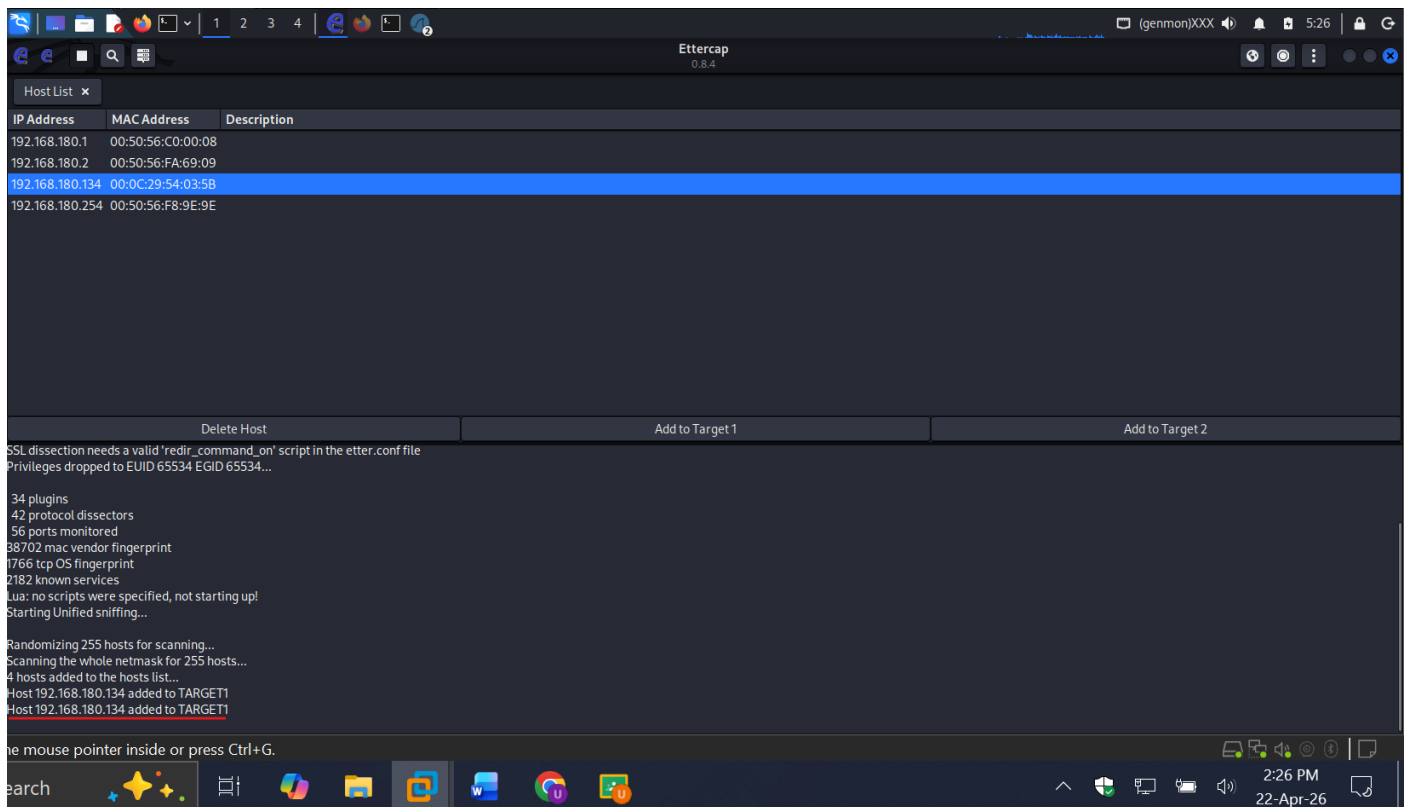
Interface Selection: The Ettercap tool is launched on a Linux system to begin a network sniffing session on the primary eth0 interface.



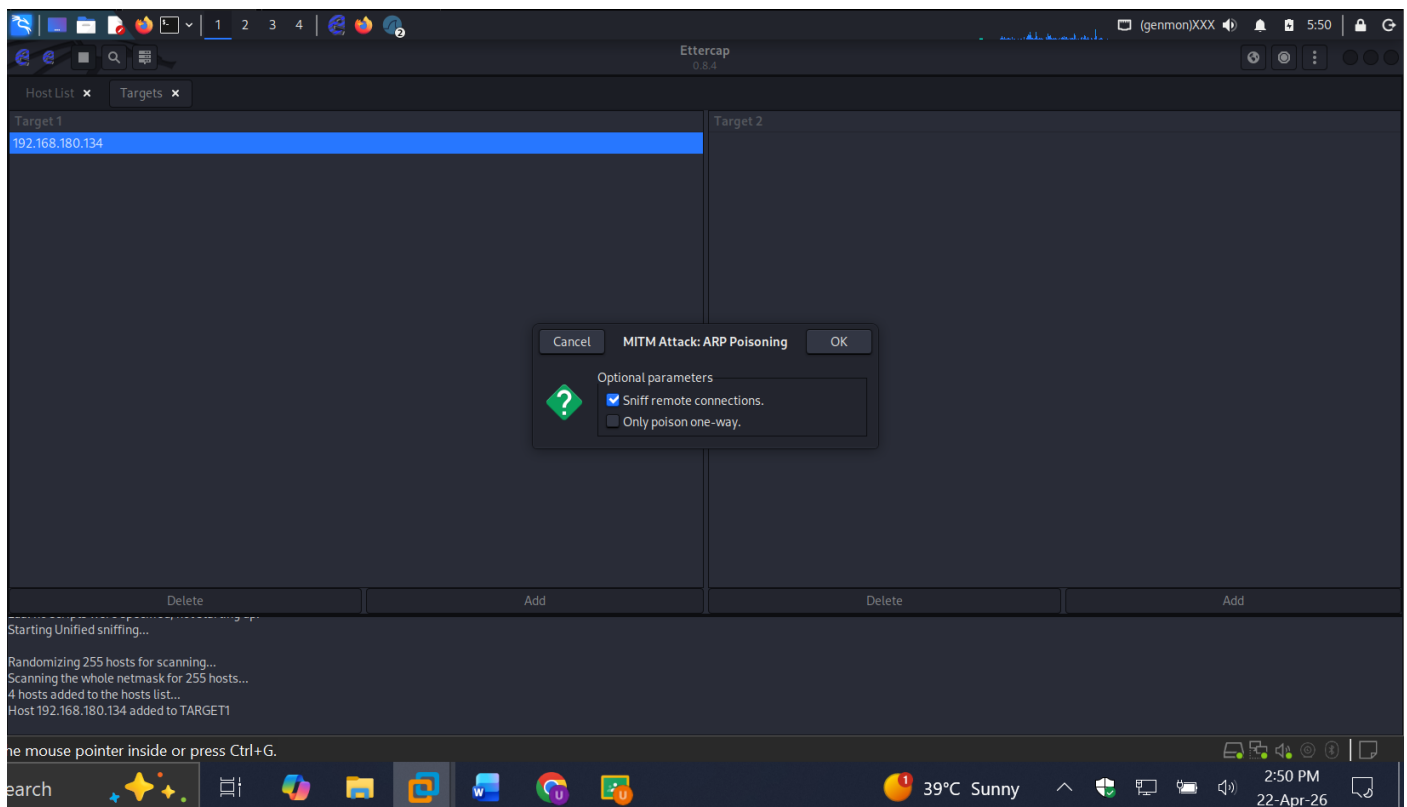
Scan for Host: The tool performs a network scan to discover active hosts, identifying their IP and MAC addresses to determine potential targets for interception.



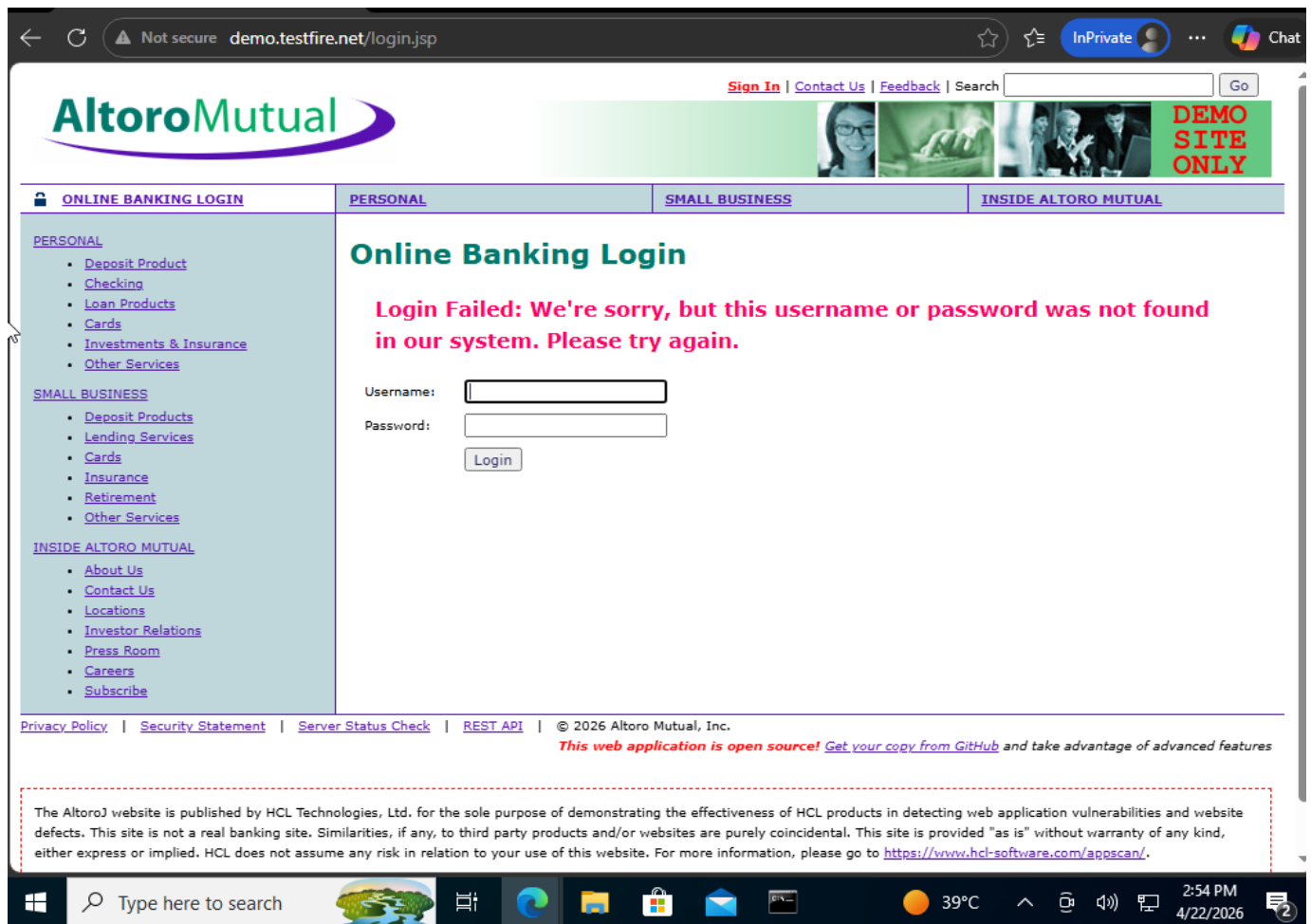
Add to Target 1: A specific host (192.168.180.134) is selected and added to Target 1 to focus the interception efforts on that particular device.



MITM Attack ARP Poisoning: An ARP Poisoning attack is initiated, which deceives the target's ARP cache to redirect its network traffic through the attacker's machine.



Browsing at Target: A user attempts to log in to an insecure, unencrypted HTTP banking website (demo.testfire.net), which transmits all data in cleartext.



Credentials Captured: Ettercap successfully captures the transmitted HTTP POST data, revealing the victim's username (userUsama) and password (user247141) in plain text.

